STM32C5A3 金手指核心板



MCU、晶体、复位、USB、电源及状态灯保留在 Core

金手指自定义针序：MCU 下半接正面 B 排，上半接背面 A 排

四线 SWD 座在 Expansion；主控仍为 C5A3，不照搬 C551 启动与时钟电路

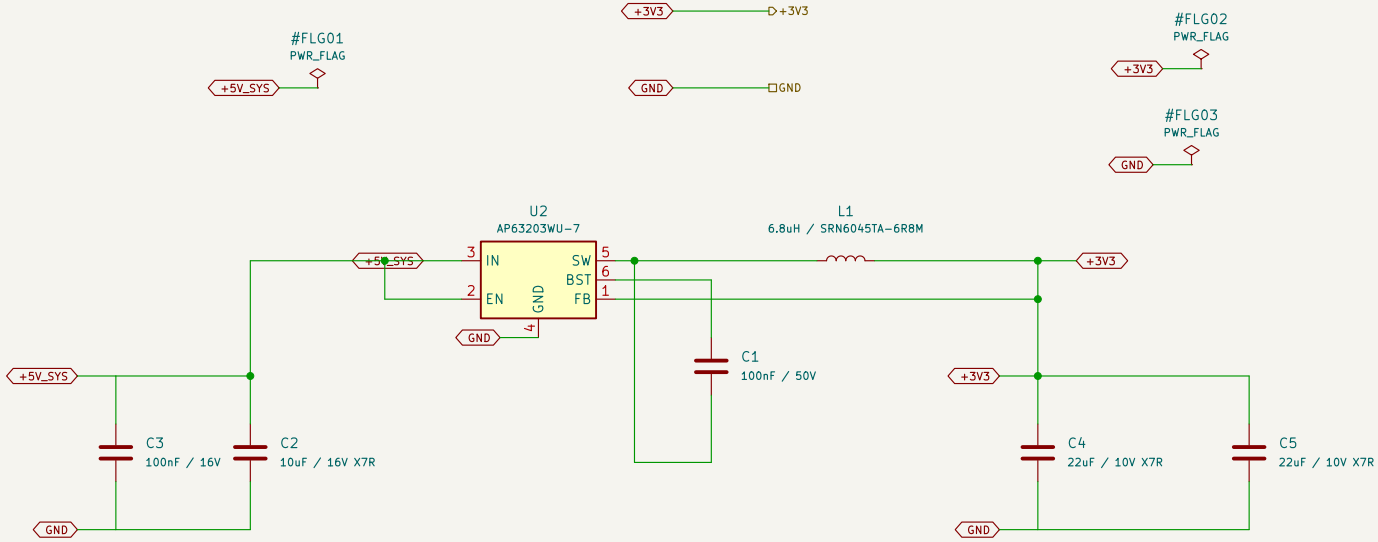
CoralC5-Core

Sheet: /
File: CoralC5-Core.kicad_sch

Title: CoralC5-Core 模块总览

Size: A4	Date: 2026-10-23	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 1/4

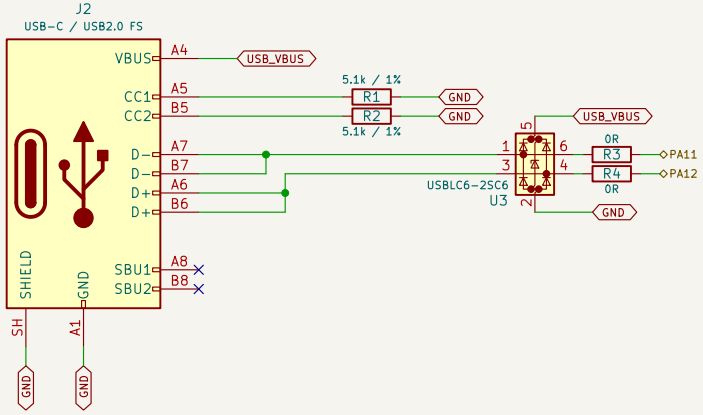
五伏从金手指输入；JP1 已移除，USB 五伏直接接 D2 阳极

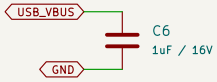


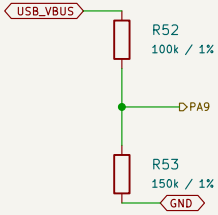
输入、输出电容采用较大封装以保留直流偏压后的有效容量

电感饱和电流大于峰值限流，反馈端直接采样三点三伏









整机主电源接 Expansion；USB 经 D2 供电，仅供低负载调试

Core 的三点三伏输出只供本板与小负载参考

布线复查版本：剩余问题见 DRC 记录

CoralC5-Core

Sheet: /电源输入、三点三伏降压与原生 USB/

File: 02_Power_USB.kicad_sch

Title: 电源输入、三点三伏降压与原生 USB

Size: A4	Date: 2026-10-23	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 2/4

主控 STM32C5A3VGT6：所有可用 GPIO 均引出至扩展页

U1A
STM32C5A3VGT6
PA0 23
PA1 24
PA2 25
PA3 26
PA4 29
PA5 30
PA6 31
PA7 32
PA8 67
PA9 68
PA10 69
PA11 70
PA12 71
PA13 72
PA14 76
PA15 77

U1B
STM32C5A3VGT6
PB0 35
PB1 36
PB2 37
PB3 89
PB4 90
PB5 91
PB6 92
PB7 93
PB8 95
PB9 96
PB10 47
PB12 51
PB13 52
PB14 53
PB15 54

U1C
STM32C5A3VGT6
PC0 15
PC1 16
PC2 17
PC3 18
PC4 33
PC5 34
PC6 63
PC7 64
PC8 65
PC9 66
PC10 78
PC11 79
PC12 80
PC13 7

U1D
STM32C5A3VGT6
PD0 81
PD1 82
PD2 83
PD3 84
PD4 85
PD5 86
PD6 87
PD7 88
PD8 55
PD9 56
PD10 57
PD11 58
PD12 59
PD13 60
PD14 61
PD15 62

U1E
STM32C5A3VGT6
PE0 97
PE1 98
PE2 1
PE3 2
PE4 3
PE5 4
PE6 5
PE7 38
PE8 39
PE9 40
PE10 41
PE11 42
PE12 43
PE13 44
PE14 45
PE15 46

U1G
STM32C5A3VGT6
+3V3 → VDD 11, 28, 50, 75, 100
VSS 10, 27, 49, 74, 99
VREF+ 21
NRST 14
VREF− 20
VCAP 48

0 / VREF LINK
+3V3 — R5 — VREF_MCU
NRST

C7
2.2uF / 6.3V X7R
GND

+3V3 — R6 10k — C16 100nF / 16V — GND

SW1
RESET / SD-1107 / 3x6x2.5
GND

C8 100nF / 16V
C9 100nF / 16V
C10 100nF / 16V
C11 100nF / 16V
C12 100nF / 16V
C13 4.7uF / 10V

VREF_MCU
C14 1uF / 10V
C15 100nF / 16V
GND

#FLG04
PWR_FLAG
VREF_MCU

Y1
24MHz / CL18pF
PH0 — 1 — 3 — PH1
2 — GND

C17 27pF / C0G — GND
C18 27pF / C0G — GND

U1F
STM32C5A3VGT6
PH0 12
PH1 13
PH3 73
PH4 19
PH5 22
PH15 6
PC14 8
PC15 9
PH2_/_BOOT0 94

SW2
BOOT0 / SD-1107 / 3x6x2.5
PH2 — +3V3
R7 10k / BOOT0
GND

+3V3 +3V3
GND GND
PA11 PA11
PA12 PA12
PA9 PA9
PA1 PA1
PA2 PA2
PC1 PC1
PA7 PA7
PC4 PC4
PC5 PC5
PB12 PB12
PB13 PB13
PB15 PB15
PC10 PC10
PC11 PC11
PC12 PC12
PD8 PD8
PD9 PD9
PA13 PA13
PA14 PA14

五组供电引脚各配一颗去耦；VCAP 只接电容

已取消 LSE 晶体；PC14/PC15 直连金手指；HSE 负载电容首板测频微调

布线复查版本：剩余问题见 DRC 记录

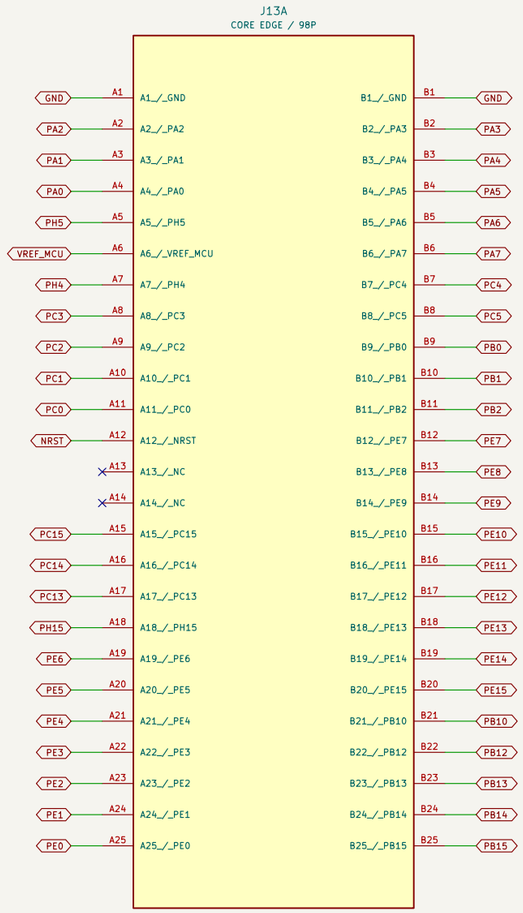
CoralC5-Core

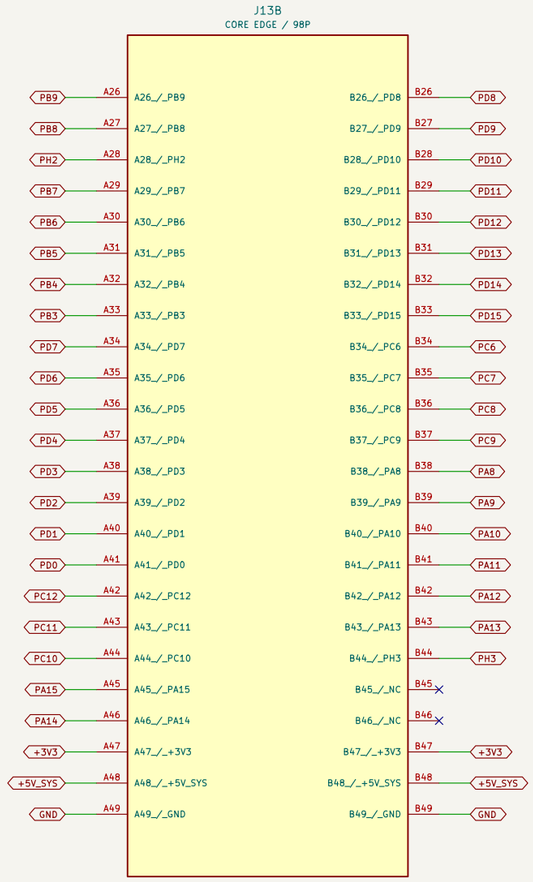
Sheet: /主控端口、供电、复位与时钟/
File: 03_MCU.kicad_sch

Title: 主控端口、供电、复位与时钟

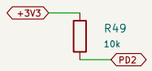
Size: A3	Date: 2026-10-23	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 3/4

Coral-C5 自定义针序：A 背面接 MCU 上半，B 正面接 MCU 下半；严禁插入电脑插槽

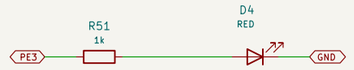


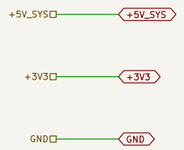












四线 XH 调试座位于 Expansion；Core 调试测试点已移除

PH0/PH1 仅接 HSE：PC14/PC15 直连金手指

布线复查版本：剩余问题见 DRC 记录

CoralC5-Core

Sheet: /公共金手指与核心板操作/

File: 04_Interface.kicad_sch

Title: 公共金手指与核心板操作

Size: A3	Date: 2026-10-23	**Rev: B1**
KiCad E.D.A. 10.0.6		Id: 4/4